Arundhati Balasubramaniam

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Highly motivated Master's student at EPFL and ETH with a background in digital humanities and computer science, specialising in language models (LLMs) and software engineering. Looking for full-time opportunities starting in Switzerland June 2026.

EDUCATION

Eidgenössische Technische Hochschule (ETH) Zurich, Switzerland *November 2025 – Present*
Master's Thesis, Interactive Visualisation & Intelligence Augmentation Lab
Topic: Evaluating Reversibility and Semantic Drift in the Low-Resource Round-Trip-Translations

École polytechnique fédérale de Lausanne (EPFL), Switzerland *September 2023 – May 2025*
Master's Candidate, College of Digital Humanities, *GPA*: 5.15/6.0
Relevant Coursework: Machine Learning; Applied Data Analysis; Data Visualisation; Computational Social Media

Ashoka University, India *September 2020 – May 2023*
Bachelor's in Computer Science (Honours), *GPA*: 3.88/4.0
Awards and Honours: Magna Cum Laude, Bronze Medal in CS, Computer Science Representative, Dean's List (5/6)

WORK EXPERIENCE

Google London | *Software Engineering Intern* *July 2025 – October 2025*
- Designed and implemented a Vertical Stacked Bar Chart component for the Foundations team of Fitbit.
- Developed UX specs for the static and animated chart in Figma, implemented a three layer abstracted public component for use by internal feature teams in Kotlin for Android devices.

Effixis | *LLM Engineer Intern* *February 2025 – June 2025*
- Implemented a streamlit app for a Swiss conglomerate to visualise supply chain scenarios, including tariffs, FX analysis and EBITs.
- Designed 2 React Tailwind apps to implement bespoke RAG LLMs for a Swiss manufacturing company.

EPFL Digital and Cognitive Musicology Lab | *Student Researcher* *May 2024 – August 2024*
- Designing and implementing a novel visualisation technique in a React app to better indicate tonal hierarchy in sheet music for analysis under supervision of Latour Prof. Martin Rohrmeier.
- Visualising non-conflict notation for Schenkerian analysis using SigmaJS, Verovio on MEI documents.

Centre for Analysis of Networks Data and Insight Derivation | *Co-Founder, Researcher May 2022 – August 2022*
- Performed sentiment, frequency and network analysis on over 500,000 tweets about the Russia-Ukraine War
- Computed and visualised topic and hashtag absenteeism in 8 major Indian media houses' headlines and articles

PROJECTS

Botticelli| EPFL LNCO | 2024-25
- Developing LLM conversational agents with social cues like humor and empathy for neurocognitive interviews, and building a full-stack ElectronJS app to modularise these cues.

UnConstitutional AI: Reinforcement Learning to Ethically Train LLMs | EPFL DHLAB | 2023
- Replicated Anthropic's paper to build an LLM that is trained using a critique-revision loop and rewarded with reinforcement learning to provide ethical and unethical answers on command.

Finetuning MEDIAR for Yeast Cell Microscopy Segmentation| EPFL LPBS | 2023
- Applied MEDIAR to a novel, LBPS-curated yeast cell microscopy dataset (YeaZ) which resulted in a fine-tuned model that achieves an F1- score of 0.977 and AP score of 0.956 on the dataset.

Lume | EPFL+ECAL Lab | 2023
- Designed a high-fidelity Figma prototype for an app that can be used to gamify energy conservation in student dorms in Switzerland.

Digital Yoknapatawpha | University of Virgina | 2022-23
- Collaborated with Professor Johannes Burgers to design an interactive supplemental site for Norton's *Absalom, Absalom!* using a novel scrollytelling graph technique with Chart.JS and Scrollama for a dynamic experience.

SKILLS

Technical: Python (PyTorch, Pandas, SciKit, Streamlit), JS (React, D3, Electron, P5, Sigma), Java, C, R
Tools & Design: Docker, Figma, Adobe CC, GIS, AWS
Languages: English (C2), Hindi (C2), Tamil (C2), Spanish (A2), French (A1)